

arc_temp_humid - outstanding work

Updated 6 September 2026. The repository rename and the main-site move are complete and are no longer listed here; see `CHANGELOG.md` for what was done.

Everything described below is **not yet done**. What *is* done is in `CHANGELOG.md` ; the architecture is in `CLAUDE.md` .

1. Automate the ARC UK Omnisense fetch

This is the only item still blocked on information I do not have.

The UK export is added **by hand** to

`data/omnisense_uk/omnisense_uk_YYYYMMDD_HHMM.csv` . The timestamp is the fetch time and is what the sidebar's freshness note reads, so keep the format.

`fetch_omnisense.py` is already parameterised by site. To switch it on:

1. Fill in the commented-out `SITES["uk"]` entry, in particular `site_nbr` .
2. Add a `--site uk` step to `.github/workflows/update-dashboard-data.yml` , next to the existing Omnisense step, with the same `continue-on-error` .

Finding the site number

It is not in the CSV: the export carries site *names*, not ids, and the `download_935909450.csv` filename was a download-job id.

Log in at omnisense.com, select the site, and read `siteNbr=` from the address bar. That is where the Tanzanian `152865` came from - it appears in `fetch_omnisense.py` as `dnld_rqst.asp?siteNbr=152865` .

The site is named "**Simmonds.Mills Retrofits**" - the `site_name` on every block of the UK export.

Credentials are probably already correct

The UK export contains a sensor labelled `House 5 Metal Roof S.E` alongside the Grove, Holywell, Chestnuts and No. 59 sensors, so one login already sees both the Tanzanian and UK material. The existing `OMNISENSE_USERNAME` / `OMNISENSE_PASSWORD` secrets will very likely work, with only `site_nbr` differing.

If not, `SITES["uk"]` accepts `username_env` / `password_env` naming its own secrets, falling back to the shared pair when absent.

When it runs

The UK export is one CSV covering several sites - Grove, Holywell, Chestnuts, No. 59 - of which this dashboard uses four sensors. `GROVE_SENSORS` and `HOLYWELL_SENSORS` in `build.py` filter by sensor id, so the rest are ignored rather than plotted. Those two sets are what to update if sensors change.

2. Decide the UK overheating threshold

Currently **none**: the three UK datasets have `"threshold": None`, so no red band is drawn and the sidebar control is hidden. Tanzania keeps 32-35 C.

Having now read CIBSE TM59 (2017), the position is clearer than "pick a number", and leaving it off may well be the right answer.

What TM59 actually requires

Compliance for a naturally ventilated home is **both** of (section 4.2):

- **(a) Living rooms, kitchens and bedrooms.** The hours where dT is at least 1 K, May to September, must be no more than 3% of occupied hours. This is TM52 Criterion 1, and dT is measured against the **adaptive** comfort limit, not a fixed temperature.
- **(b) Bedrooms only.** Operative temperature between 22:00 and 07:00 must not exceed **26 C** for more than 1% of annual hours. TM59 spells out the count: 32 hours, so 33 or more is a fail.

So the only fixed number in TM59 is 26 C, and it applies **to bedrooms, at night, counted over the year**. For a living room or kitchen there is no fixed threshold at all - the test is the adaptive one, which the adaptive comfort chart already serves.

What that means for the two UK sensors

Logger	Name	Room type	Fixed threshold applies?
169502D1	Living Room (No. 57), Grove	living room	no - criterion (a) only
0E3C12EC	Internal Ambient, Holywell	unknown	only if it is a bedroom

Grove's sensor is a living room, so TM59 gives it no fixed band. Holywell's is labelled only "Internal Ambient"; **that is the room type worth confirming.**

Three honest options

1. **Leave the threshold off** (current state). Defensible: neither sensor is a confirmed bedroom, and criterion (a) is already covered by the adaptive chart.
2. **Add 26 C only where a logger is a bedroom.** Correct, but the threshold is currently per-region, so this needs a per-logger threshold - a real change, though not a large one.
3. **Draw 26 C across the UK region as a rough reference.** Quickest, and the least honest: TM59's 26 C is night-only and bedroom-only, so a permanent band across a 24-hour living-room chart misrepresents the standard.

Two caveats before quoting TM59 numbers from this dashboard

- TM59 is a **design and modelling** methodology. It assumes standard occupancy profiles and dynamic simulation, and evaluates **operative** temperature. This dashboard plots **measured air** temperature, which is already listed as a known limitation in `adaptivecomfort.md`. Numbers taken from here are indicative of TM59, not a TM59 assessment.
- Criterion (a) is adaptive against **TM52 Category II**, whose upper limit is $0.33 \times T_{rm} + 18.8 + 3$. The dashboard currently offers ASHRAE 55 and the Vellei models, neither of which is that line. If the UK datasets are ever to be read against TM52/TM59, adding a TM52 Cat II band to the comfort model dropdown is the change that would do it - and is probably worth more than the threshold band.

3. Confirm the UK Open-Meteo coordinates

Working and live, on one assumption worth correcting.

Both UK buildings have their own Open-Meteo feed, fetched daily. Each drives its own building's adaptive comfort running mean and both appear on the ARC UK chart. The coordinates in `fetch_openmeteo.py` are **town-centre positions**:

```
"grove": {"lat": 52.0567, "lon": -2.7160, ...} # Hereford
"holywell": {"lat": 52.9186, "lon": -4.2372, ...} # Criccieth
```

How much the precision matters

Measured, not assumed. Open-Meteo snaps a request to its model grid, and that grid is finer than the 11 km often quoted. Comparing the returned hourly series against the town-centre baseline, June to August 2026 (2208 hours):

Offset from baseline	Mean difference	Max	Hours differing by >0.5 C
~0.5 km	0.18 C	1.6 C	4.9%
~2 km	0.51 C	4.0 C	32.8%
~9 km	1.52 C	4.9 C	95.6%
~20 km	1.63 C	8.0 C	83.0%

So getting within roughly half a kilometre is worth doing; beyond about two kilometres the series starts to drift meaningfully from the site.

The privacy point

These are private homes and **this repository is public**, so an exact coordinate committed here is effectively a published home location.

Half a kilometre of precision is enough, and that is about **two decimal places**: 0.01 degrees of latitude is around 1.1 km, and of longitude at these latitudes around 0.7 km. So the resolution to commit is two decimals - it puts the request in the right grid cell without pinpointing a dwelling.

If exact positions or postcodes are supplied, round them to two decimals before they go in the file. It is a one-line change per site and the next daily run picks it up.

As a check that the two feeds are genuinely distinct: Grove averages 1.03 C warmer than Holywell across 30 497 paired hours, only 611 of them identical - consistent with inland

Herefordshire against coastal North Wales.

4. Known limitation, not a task

ASHRAE 55 Section 5.4.1(a) requires that no heating is in operation. Nothing at any site records heating status, and nothing in a temperature and humidity trace reliably separates a heated room from a merely warm one.

The dashboard states this rather than inferring it: a bolded line in the sidebar applicability panel directly beneath the quoted criterion, and in the tooltips explaining the method. The 10-33.5 C running-mean gate is enforced automatically and greys out readings the standard does not cover.

That gate now has teeth it did not have before, because the UK Open-Meteo feeds reach back to March 2023: once UK winter indoor data exists, cold-weather readings will be greyed and excluded from the comfort percentage. But **a mild day with the heating on will still pass every test the dashboard can apply.**

Closing this properly needs a heating-status input, not a cleverer algorithm. See `adaptivecomfort.md` section 2 for the full reasoning, including why calendar-month filtering was considered and rejected.